

Lesson 6-5 · Quick

Properties of Logarithms + DOK-3 Power
Property Proof & Richter Generalization

Klimsara-adapted · Student-centered · No Blooket

Do Now — Log Properties Warm-Up

Algebra 2 .

Lesson 6-5 · Quick

DOK 1–2

7 min

bridge to log properties launch

3 items: error analysis + expand + condense

1. **Error Analysis** A student approximates $\log_2 9$ by entering $\frac{\ln 2}{\ln 9}$ on her calculator. Why is this wrong?
2. **Expand:** $\log_6 \frac{49}{5}$
3. **Condense:** $5 \ln s + 6 \ln t$

Launch — Example 3: Condense Into a Single Log

Algebra 2 · Lesson 6-5 · Quick

DOK 1–2

5 min

teacher models

Rules Card (keep visible all period) **PRODUCT:**

$$\log_b(MN) = \log_b M + \log_b N$$

QUOTIENT: $\log_b(M/N) = \log_b M - \log_b N$

POWER: $\log_b(M^n) = n \cdot \log_b M$

CHANGE OF BASE: $\log_b x = (\log x)/(\log b)$

Condensing order: POWER first → then PRODUCT / QUOTIENT

Teacher models: $4 \log_4 m + 3 \log_4 n - \log_4 p$ Step 1

Explore — TPS + DOK-3 Paired Driver Algebra 2 .

Lesson 6-5 · Quick

DOK 2–3

28 min

student work

6 items in order. Plan ~12 min for #13 + #40. Wed-F: skip #3.

Practice #19 · Condense $\log_5 6 + \frac{1}{2} \log_5 y$

Practice #21 · Condense $\frac{1}{3} \ln 27 - 3 \ln(2y)$

Practice #34 · Solve (change of base) $7^x = 100$

DOK 2

5 min

DOK-3 reveal + Topic 6 bridge

Sentence frame · Power Property Proof

(#13) “Let $x = \log_b M$. Then $b^x = M$. Raise both sides to n : $(b^x)^n = M^n$, so $b^{(nx)} = M^n$. By the definition of log, $\log_b(M^n) = nx = n \cdot \log_b M$.”

Sentence frame · Richter (#40 Part C)

$R(150A_0) - R(A_0) = \log(150A_0/S) - \log(A_0/S) = \log 150 \approx 2.18$.

The larger earthquake is about **2.18 Richter units** greater.

Exit Ticket — Two-Part Summary

Algebra 2 · Lesson

6-5 . Quick

DOK 1–2

5 min

not graded

Complete on your exit ticket A biologist models a population as $P = 3 \log t + \log 4$.

1. Write P as a single logarithm: $P = \log(\text{_____})$
2. Solve for t when $P = 2$: $t = \text{_____}$
3. One biggest thing I learned today:
